

Technical Document #629: Cloud Computing - Comprehensive Overview

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Microservices architecture structures applications as a collection of loosely coupled services. Each service runs in its own process and communicates via APIs.

Containerization packages applications with their dependencies. Docker creates lightweight, portable containers that run consistently across environments.

Platform as a Service (PaaS) provides a platform for developing and deploying applications. Heroku and Google App Engine are PaaS examples.

Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files. Terraform and CloudFormation are popular IaC tools.

Content delivery networks (CDNs) distribute content globally to reduce latency. CloudFront and Cloudflare are popular CDN providers.

Multi-cloud strategies use services from multiple cloud providers. They reduce vendor lock-in and optimize for cost and performance.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Key Takeaways:

- Serverless computing allows developers to run code without managing servers.
- Kubernetes orchestrates containerized applications across clusters of machines.
- Infrastructure as a Service (IaaS) provides virtualized computing resources over the internet.